What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to compare your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

system: the object or objects chosen for analysis

01 Bring back an earlier idea

An ideal asteroid orbit has nearest distance 3 AU and farthest distance 6 AU. Where is it fastest?
Explain with the equal-area law.

Word help: orbit = a path around another body under gravity

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A 2 kg cart and a 4 kg cart move right at 3 m/s. Their momenta mv are 6 and 12 kg·m/s. Stopping the second requires twice the momentum change.

Your turn: Draw the velocity and momentum arrows for both carts.
Explain which pair has equal lengths.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

Carts A and B both move right at 2 m/s. A has mass 1 kg; B has mass 3 kg. Which has more momentum?
State the controlled quantity.

Sentence starter: I notice … . This supports … because … .



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Build the solution

Both carts stop in 3 s.
Use force size=momentum change size÷time.
Find each average stopping force size.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage?
Explain.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A supplied gravity model has $Gm_1m_2=144$ N·m² and separation 6 m.
Find the force and its direction.

Word help: separation = distance between the specified objects or centers

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg cart moving at 3 m/s stops in 3 s.
Find momentum change size and stopping force size.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find a lighter, faster cart with the same momentum as a 4 kg cart at 3 m/s.
Explain why equal momentum does not mean equal mass or speed.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3